

Dylan Kawalec

Head of Developer Relations | Founding Engineer & Solutions Architect | Privacy-Preserving AI & Blockchain

Confidential Compute & Privacy: SGX/TDX TEEs, zero-knowledge proofs, remote attestation, and threshold signing.

Experienced Research Engineer • Developer Platform SDK Architect • Confidential AI infrastructure, LLM agent systems, and technical developer education.

B.S. Computer Science, Western Governors University • **A.A.S. Computer Programming**, MiraCosta College

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Summary

I build and ship confidential systems: TEE-backed services (SGX/TDX), zero-knowledge authentication, and privacy-first blockchain infrastructure.

Current build focus: confidential AI runtimes on TDX/SGX with remote attestation, zero-trust verification boundaries, and verifiable execution.

Execution model: founding-engineer architecture with hands-on delivery across APIs, distributed services, Solidity smart contracts, and formal specifications.

Deep Tech Developer Relations function: documentation, reference implementations, and partner onboarding that convert architecture into shipped integrations.

Focus Areas & Tooling

Compute/Security: Intel SGX/TDX (Fortanix, dstack/TEE), Phala Cloud remote attestation, enclave TLS, MPC, threshold signing.

ZK/Cryptography: zero-knowledge authentication, commitments and Schnorr-style proofs, threshold signing, protocol and threat-model design.

Languages/Frameworks: Python (8+ years: PyTorch, Matplotlib, NumPy, SymPy, cryptography, gmpy), Rust (Tokio, Actix, wagyu, serde), Solidity, Go, Node.js/TypeScript/JavaScript.

Systems: TLA+/PlusCal specs, distributed services, API/microservice architecture, Docker/GitHub Actions/Terraform, Postgres/Redis.

Automation Tooling: LaTeX, Codex, Warp OZ agents with [Warp Engine](#), Cursor IDE, n8n workflows with [opgrok](#), CoreThink API, Sonnet 4.5.

Selected Builds (Shipped / Demonstrated)

- **Confidential terminal agents:** [DI](#) (Solidity ERC-8004 interface) and [Warp Engine](#) for attested TDX private-agent execution.
- **Decentralized front-end infrastructure:** [defe](#) Rust CLI for SGX/Fortanix provisioning, enclave TLS, and threshold signing to reduce operator trust risk.
- **Protected password entry:** [ENI6MA demo](#) spanning web, CLI, and on-chain verification flows under confidential research constraints.
- **Zero-trust verifier network:** [cosmic-hack](#) prototype for TEE-only execution and ZK-style boundary verification.
- **Learned compression R&D:** Evoformer-inspired plus zstd pipeline for deterministic, verifiable transforms over structured and unstructured binary data.

Experience

Head of Developer Relations & Solutions Architect *Phala Network*

May 2025 – Dec 2025 (FT)

Confidential compute (TEE) platform: Phala Cloud / dstack workflows

- Authored Phala Cloud + dstack documentation and reference deployment flows that moved 12+ partner teams from evaluation to production confidential-compute deployments.
- Led Confidential AI rollout by deploying private agents in **Intel TDX** via **dstack**, using remote attestation to provide execution guarantees.
- Implemented [DI](#) Solidity **ERC-8004** contracts and built [Warp Engine](#) for private terminal-agent orchestration; co-authored the **Trustless Cloud Agents** specification with the Ethereum Foundation.

- Presented dstack/TEE workflows to the Confidential Computing Consortium, onboarded partners, and supported Linux Foundation acceptance (17/17 vote).
- Grew dstack repo visibility from 34 to 400+ GitHub stars in 6 months through docs, demos, integrations, and community programs (community calls and SF Bay Area meetups with Flashbots, tee.fail, Secret Network, and others).

Founding Engineer / Principal Architect (Independent Research) *Cosmic Inc.* / Summer 2024 – Jan 2026
ENI6MA LLC

Confidential R&D: zero-trust verification, zero-knowledge authentication, deterministic verification, replay-resistant transcripts (public synopsis)

- Designed a zero-trust access architecture using trusted execution environment-only execution, remote attestation, and a cryptographic proof boundary for human and agent access.
- Built verifier prototypes in **Python** and **C++**, implementing commitment-based authentication and challenge-response verification flows with hardened transcripts.
- Wrote formal system specifications and threat models, then translated them into test harnesses and reference implementations for reproducible verification.
- Produced the engagement handoff (provisional patent package, private codebase, demos, and 24+ internal papers) and supported a **\$168K** friends-and-family raise through architecture reviews and technical demos.

Research & Development Engineer *Osmosis Foundation* Apr 2024 – Sep 2024 (Contract)

Cosmos infra R&D: decentralized front-ends, Fortanix SGX enclave deployment, validator signing/neutrality

- Built and shipped [defe](#), a production-grade **Rust CLI** that automates **Fortanix SGX** build/sign/enclave workflows for secure, reproducible decentralized front-end provisioning.
- Implemented **enclave-native ACME/Certbot TLS issuance** and **key lifecycle automation** (generation, sealing, rotation) to keep private material inside SGX while enabling public HTTPS endpoints.
- Designed **threshold-signing partitioning** for operational signing paths to reduce single-operator trust and improve neutrality guarantees for Osmosis front-end operations.
- Wrote end-to-end **SGX deployment and ops documentation** (attestation, secrets handling, edge rollout) to move Cosmos teams from prototype setups to production-style environments.

Head of Developer Relations *GOAT Network (BTC L2)* Jul 2024 – Jan 2025 (Contract)

BTC L2 enablement: docs, integrations, workshops

- Authored [GOAT Network technical documentation](#) for zkMIPS, sequencer APIs, and EVM/Solidity integration paths for BTC L2 teams.
- Delivered the [Founders Club MVP](#) and onboarded 100+ founders through structured deployment workflows.
- Built and managed **40+ ecosystem partnerships**, including major infrastructure providers listed in the [GOAT ecosystem directory](#).
- Led BTC L2 workshops at BTC Nashville and Staking Summit Singapore, then translated workshop feedback into integration runbooks.

Head of Developer Relations *Findora / Discreet Labs (now Fractal Network)* May 2022 – Oct 2022 (Contract)

Confidential Ethereum Virtual Machine Auditing & Developer Enablement + Leading ecosystem growth

- Delivering validator, bridging, and privacy workflow education for global ecosystem teams.
- Onboarded **80+ validators** into Findora node operations through direct setup support, strengthening network participation during Series A visibility.
- Supported ETHAmsterdam launch narrative during a **\$12M+** token raise window, alongside [Yellow Submarine](#) and [FairySwap ecosystem](#) rollouts.

CTO & Principal Engineer *Invo Technologies* Apr 2023 – Jan 2025 (Contract)

Substrate + privacy rail R&D; developer monetization tooling

- Built the initial [Invo platform MVP](#) for game-economy monetization and player-economy analytics.
- Designed a privacy-preserving decentralized exchange using Parity's (Substrate Blockchain Framework) for game-to-game token transfers (written in Rust), Aleo zkOS privacy rails, and other EVM interoperability requirements.
- Designed an interoperable **Token Bridge** in Solidity, creating smart contract interfaces for a multi-party Hash Time Lock Contract (MP-HTLC) settlement system. Used AES, OpenSSL, and TLS to secure CI/CD pipelines, maintain audit logs, manage revocations, and enable on-device remote attestations via Aleo [zkVM-mediated verification](#).
- Led implementation of [POEMS \(Proof of the Effective Majority Staked\)](#) — an on/off-chain Schnorr signature scheme; won Aleo Best Tech 2023 and supported a \$50K Aleo Foundation grant. Built the commitment-challenge-

response authentication flow integrated with PostgreSQL and wallet infrastructure to secure in-game INVO token operations (deposits, transfers, revocations) across Unity/Unreal wallets, with Aleo privacy interoperability to prove on-chain/in-game event validity and protect asset custody.

Independent Research Studio (contracts) *Chaotic Beauty LLC*
Single Member LLC for R&D, solutions architecture, Developer Relations contracts

Nov 2020 – Present

- Ran a contract studio for 6+ month engagements across blockchain teams, covering security architecture, integrations, and developer platform enablement.
- Delivered MPC/threshold-signing custody patterns and key-management hardening to reduce operational key exposure in production digital-asset systems.
- ZKM Partnership with Kevin Liu (CEO of ZKM) to build ZKM Documentation for the Bitcoin Layer 2 ecosystem.

Earlier: Tesla (Model 3 Technician) • Zygo (IBF Technician; precision optics supporting JWST primary-mirror metrology) • BNB Chain Labs (US Tech Evangelist; first major US hire; onboarded teams incl. StorJ & Polyhedra; host, [BNB Podcast](#)) • Harmony (Developer Captain/DAO Governor; secured \$100K DAO funding; onboarded 700+ devs) • Bridge Builders DAO (Ethereum smart contract developer/consultant).

Recognition, Credentials & Publications

Selected Awards

- **ETHGlobal SF 2024:** [Zend It](#) — Ledger Tech Award (1st), Polygon Best zkApp (1st).
- **ETHDenver 2024:** AOS *Hacker Deathmatch*.
- **VeChain × EasyA 2024:** [PowerUp](#) (2nd).
- **Aleo 2023:** POEMS (Best Tech).
- **ETHAmsterdam 2022:** [DegenBox](#) — EPNS Best Use of Frontend SDK; Boba Network Best Use.
- **KCACTF 2018:** Best Partner (Theatre) — [IMDB](#).

Credentials RareSkills: Research Fellow (May 2023) • **ZKU (Harmony ONE):** Zero Knowledge University Developer (Mar 2022)

Selected Writing, Docs & Research

- [Phala Cloud / dstack Documentation Suite](#)
- [GOAT Network Protocol Documentation](#)
- [CoinFi Research Archive](#)
- [ENI6MA Research Fellowship \(public excerpts\)](#)
- [Aleo Developer Enablement Strategy](#)
- [Churn Monetization is the Future of GameFi](#)
- [The Future of GameFi and Gaming: Invo's DLAL and the Digital Economy](#)
- [Bridging Blockchain and Gaming with Invo's Default Currency Solution](#)

Selected Talks, Podcast Hosting & Public Appearances Online

- [Host, The Blackbox Podcast](#) — Sponsored by Phala Network; independent guest interviews and deep dives across privacy, AI, and blockchain.
- [DevMos 2024: Decentralized Front Ends Workshop](#)
- [Staking Summit: GOAT Network BTC L2 Workshop](#)
- [Bread N' Butter Podcast \(BNB Chain Labs\)](#)
- [Cosmic Public Proof \(YouTube\)](#) — published a carrier-audio proof flow with permalinked verification context in the video description.